

Diagnostic Knowledge Survey

AI Automation & Agents — Pre-Instruction Baseline

12 Questions

~10 Minutes

Ungraded

⚠ **Do not look up answers before completing this survey.**

This survey captures your *current* mental models before instruction begins. Your responses form the baseline the course uses to measure learning growth. All answers are confidential and will not affect your grade.

0 of 12 answered

SECTION 1 — PRIOR EXPERIENCE WITH AUTOMATION TOOLS

QUESTION 1 OF 12

Which of the following automation tools or platforms have you used before?
(Select all that apply.)

☐ Zapier / Make (Integromat)

☐ UiPath, Automation Anywhere, or another RPA platform

☐ Excel macros or Google Apps Script

☐ Shell scripts (Bash, PowerShell)

☐ Python scripts for task automation

☐ None of the above

QUESTION 2 OF 12

How many years of experience do you have using any form of workflow automation in a professional or academic setting?

☐ None — this is new to me

☐ Less than 1 year

☐ 1–2 years

☐ 3–5 years

☐ More than 5 years

QUESTION 3 OF 12

Which best describes the most complex automation workflow you have built or maintained?

☐ I have not built one

☐ A single-step trigger → action (e.g., "email arrives → save attachment")

☐ A multi-step pipeline with conditional branching

☐ A pipeline integrating multiple external APIs or services

☐ An autonomous agent or system that makes decisions without predefined rules

SECTION 2 — CURRENT WORKING DEFINITION OF 'AI AGENT'

QUESTION 4 OF 12

In your own words, what is an "AI agent"? Write 1–3 sentences. (There is no correct answer — describe what you currently believe.)

Type your definition here...

QUESTION 5 OF 12

Which of the following properties do you believe are essential to classify a system as an "AI agent"? (Select all that apply.)

- ☐ Perception — the ability to observe an environment
- ☐ Goal-directedness — the system pursues an objective
- ☐ Memory — the ability to retain state across interactions
- ☐ Tool use — the ability to call external functions or APIs
- ☐ Use of a large language model (LLM) as a core component
- ☐ Autonomy — the ability to act without step-by-step human instructions
- ☐ I am not sure

QUESTION 6 OF 12

How would you distinguish an "AI agent" from a standard chatbot or conversational AI?

- ☐ They are essentially the same thing
- ☐ Agents can take actions in the world; chatbots only generate text responses
- ☐ Agents are smarter / use a more powerful model
- ☐ Agents operate in a perception–reasoning–action loop; chatbots respond reactively to a single prompt
- ☐ I do not know

QUESTION 7 OF 12

Which of the following is the best example of a multi-agent system?

- ☐ A user and ChatGPT having a back-and-forth conversation
- ☐ A data pipeline that runs ETL jobs nightly
- ☐ A planner agent that breaks a task into subtasks and delegates them to specialist agents
- ☐ A retrieval-augmented generation (RAG) system that answers questions over documents
- ☐ I do not know

SECTION 3 — COMFORT LEVEL WITH PYTHON

QUESTION 8 OF 12

Rate your overall Python programming proficiency.

1

2

3

4

5

QUESTION 9 OF 12

Which Python concepts are you comfortable using independently? (Select all that apply.)

- ☐ Variables, loops, conditionals, and functions
- ☐ Object-oriented programming (classes, inheritance)
- ☐ Calling external APIs using requests or httpx
- ☐ Asynchronous programming (asyncio, await)
- ☐ Data manipulation with pandas or numpy
- ☐ LLM libraries (LangChain, LlamaIndex, OpenAI SDK, etc.)
- ☐ None of the above

QUESTION 10 OF 12

How comfortable are you reading and debugging unfamiliar Python code written by others?

- ☐ Not comfortable — I struggle to follow code I did not write
- ☐ Somewhat comfortable with straightforward scripts
- ☐ Comfortable with moderately complex code
- ☐ Very comfortable — I regularly review others' code

QUESTION 11 OF 12

Rate your confidence in using Python specifically for building AI or LLM-powered applications.

1

2

3

4

5

No confidence

Fully confident

SECTION 4 — OPEN-ENDED REFLECTION

QUESTION 12 OF 12

Name one repetitive task at work or school that you wish you could hand off. Describe it briefly: What triggers it? What steps does it involve? What would "done" look like?

Describe the task here...

Submit Survey

All responses are saved locally in your browser.